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Foreign object detection in a wireless power transfer system

Abstract

This disclosure provides systems, methods and apparatuses for foreign object detection (FOD) in a wireless power transfer (WPT) system. Some implementations relate generally to the use of detection coils that are excited to measure and compare a differential current through a coil pair that includes at least two detection coils. A foreign object may cause a change in impedance for one or more detection coils compared to one or more other detection coils. By detecting the differential current of the coil pair, a detection apparatus may determine that a foreign object is in proximity to one of the detection coils of the coil pair. This disclosure provides several options for the design, construction, layout, and operations of detection coils to improve foreign object detection.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This Patent Application is a National Stage of International Application No. PCT/US2022/016543, filed Feb. 16, 2022, and claims the benefit of priority to India Non-Provisional patent application No. 202111006541, filed Feb. 16, 2021, entitled “FOREIGN OBJECT DETECTION IN A WIRELESS POWER TRANSFER SYSTEM,” and assigned to the assignee hereof, the disclosures of which are incorporated by reference in this Patent Application.

TECHNICAL FIELD

(2) This disclosure relates generally to wireless power. More specifically, this application relates to foreign object detection in a wireless power transfer system.

DESCRIPTION OF RELATED TECHNOLOGY

(3) Technology has been developed to enable the wireless transmission of power from a wireless power transmission apparatus to a wireless power reception apparatus. Examples of a wireless power reception apparatus may include some types of mobile devices, small electronic devices, computers, tablets, gadgets, appliances (such as cordless blenders, kettles, or mixers), and some types of larger electronic devices, among other examples. Wireless power transmission may be referred to as a contactless power transmission or a non-contact power transmission. The wireless power may be transferred using inductive coupling or resonant coupling between a primary coil of the wireless power transmission apparatus and a secondary coil of the wireless power reception apparatus. For example, a wireless power transmission apparatus may include a primary coil that produces an electromagnetic field. The electromagnetic field may induce an electromotive force in a secondary coil of a wireless power reception apparatus when the secondary coil is placed in proximity to the primary coil. In this configuration, the electromagnetic field may wirelessly transfer power to the secondary coil.

(4) In a wireless power transfer system, when a foreign metal object (such as a key, a coin, a metallic can, or aluminum foil, among other examples) is in proximity of the electromagnetic field, the foreign metal object may be undesirably heated up due to eddy currents. This may result in safety hazards, such as fire safety hazards. Furthermore, the efficiency of wireless power transfer process may be inadvertently affected or disrupted. Traditional techniques for detecting foreign objects in a wireless power transfer system may be inadequate or ineffective to prevent such safety hazards.

SUMMARY

(5) The systems, methods, and apparatuses of this disclosure each have several innovative aspects, no single one of which is solely responsible for the desirable attributes disclosed herein.

(6) One innovative aspect of the subject matter described in this disclosure can be implemented as a detection apparatus of a wireless power transfer (WPT) system. The detection apparatus may include a plurality of detection coils including at least a first detection coil and a second detection coil. The detection apparatus may include a first driver configured to concurrently excite the first detection coil and the second detection coil during a first foreign object detection (FOD) period. The detection apparatus may include a differential current sensing apparatus configured to detect a first differential current associated with the first detection coil and the second detection coil during the FOD period. The detection apparatus may include a control unit configured to generate a foreign object detection signal based, at least in part, on the first differential current.

(7) Another innovative aspect of the subject matter described in this disclosure can be implemented as a detection apparatus of a WPT system. The detection apparatus may include a plurality of

detection coils arranged to form a FOD scan area that is at least a threshold size larger than a potential combined surface area of a plurality of power transfer coils of a WPT system. The detection apparatus may include a control unit configured to generate a foreign object detection signal based, at least in part, on a detection of a foreign object in the FOD scan area.

(8) Another innovative aspect of the subject matter described in this disclosure can be implemented as a WPT system. The WPT system may include a wireless power transmission apparatus including at least one primary coil and a first plurality of detection coils arranged to form a first FOD scan area that is at least a first size larger than a size of the primary coil. The WPT system may include a wireless power reception apparatus including at least one secondary coil and a second plurality of detection coils arranged to form a second FOD scan area that is at least a second size larger than a size of the secondary coil.

(9) Another innovative aspect of the subject matter described in this disclosure can be implemented as a method of a detection apparatus of a WPT system. The method may include concurrently exciting at least a first detection coil and a second detection coil during a first FOD period. The method may include detecting, by a differential current sensing apparatus, a first differential current associated with the first detection coil and the second detection coil during the FOD period. The method may include generating a foreign object detection signal based, at least in part, on the first differential current.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Details of one or more implementations of the subject matter described in this disclosure are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages will become apparent from the description, the drawings, and the claims. Note that the relative dimensions of the following figures may not be drawn to scale.

(2) FIG. 1 shows a block diagram of an example wireless power transfer system.

(3) FIG. 2 shows a block diagram of an example detection apparatus in a wireless power transfer system.

(4) FIG. 3 shows a block diagram of an example detection apparatus configured to excite a pair of detection coils to measure a differential current.

(5) FIG. 4 shows a chart with example magnitudes of differential currents.

(6) FIG. 5 shows a block diagram of an example detection apparatus based on a detection voltage induced by a differential current.

(7) FIG. 6 shows a circuit diagram of an example electrical configuration of a coil pair.

(8) FIG. 7 shows a circuit diagram of an example electrical configuration of multiple coil pairs.

(9) FIG. 8A shows an example timing diagram for a detection apparatus having multiple coil pairs.

(10) FIG. 8B shows another example timing diagram for a detection apparatus having multiple coil pairs.

(11) FIG. 8C shows another example timing diagram for a detection apparatus having multiple coil pairs.

(12) FIG. 9A shows a diagram of multiple coil pairs in associated detection zones.

(13) FIG. 9B shows example detection of a foreign object using the multiple coil pairs of FIG. 9A.

(14) FIG. 10 shows another diagram of multiple coil pairs in associated detection zones.

(15) FIG. 11 shows an example line diagram of detection coils.

(16) FIG. 12 shows an example line diagrams of detection coils with features to reduce the effect of a power transfer coil field.

(17) FIG. 13 shows a diagram of multiple coil pairs capable of distinguishing between movement of a wireless power reception apparatus or a foreign object.

(18) FIG. 14 shows example charts comparing detection voltages induced by differential currents of coil pairs as a result of movement of a wireless power reception apparatus or a foreign object.

(19) FIG. 15 shows a block diagram of an example detection apparatus configured to adapt detection voltages of coil pairs based on movement of a wireless power reception apparatus.

(20) FIG. 16 shows a pictorial diagram of example foreign object detection scan areas.

(21) FIG. 17 shows a flowchart diagram of an example process for detecting a foreign object in accordance with some implementations.

(22) FIG. 18 shows a block diagram of an example apparatus for use in wireless power transfer system.

(23) Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

(24) A wireless power transfer (WPT) system may include a wireless power transmission apparatus and a wireless power reception apparatus. The wireless power transmission apparatus may include one or more primary coils that transmit wireless energy (as a wireless power signal) to one or more corresponding secondary coils in the wireless power reception apparatus. A primary coil refers to a source of wireless energy (such as inductive or magnetic resonant energy producing an electromagnetic field) in the wireless power transmission apparatus. A secondary coil located in the wireless power reception apparatus may receive the wireless energy via the electromagnetic field. Occasionally, a foreign object (sometimes referred to as a foreign metal object) may be in proximity of the electromagnetic field. A foreign object may be any object that is electrically conductive or have detectable magnetic permeability and that is not part of a WPT system but is inadvertently present in an operative environment of the WPT system. Non-limiting examples of foreign objects may include a ferrous object, a metallic can, a coin, a metal spoon, a key, aluminum foil, or other electrically conductive or ferrous objects. When a foreign object is in proximity of the electromagnetic field, the foreign object may negatively impact the wireless power transfer or may become undesirably heated up due to eddy currents.

(25) There are a variety of techniques for detecting foreign objects in wireless power transfer systems. Some techniques may include detecting presence of a foreign object based on a variation in frequency of the current in a primary coil, detecting imbalanced disparities in current and voltages in the primary coil, power loss accounting based on measurements of power drawn from the primary coil, and the like. In some of the detection techniques there may be some delay in detection of the foreign objects after the wireless power transfer is initiated. The foreign object may absorb energy from a wireless power signal during this period which may result in wastage of power or unsafe heating. As WPT systems are being developed for higher amounts of power, there is an increased potential risk of foreign objects rapidly heating to unsafe temperatures.

(26) This disclosure provides systems, methods and apparatuses for foreign object detection (FOD) in a wireless power transfer system. Some implementations relate generally to the use of detection coils in a detection apparatus (such as a detection mat or integrated with component of the WPT system). A pair of detection coils may be excited during an FOD period to measure and compare a differential current through the pair of detection coils. The differential current may be a result of a difference in impedance associated with one detection coil as a result of the presence of the foreign object in the operative environment of the WPT system. For example, the foreign object may cause a change in impedance for one or more detection coils compared to one or more other detection coils. As a result of the different impedance, the various detection coils may draw different amounts of current when they are energized. For brevity, this disclosure includes a description of a coil pair that includes at least two detection coils. The difference in the amounts of current associated with detection coils of a coil pair may be referred to as a differential current. By detecting the differential current of the coil pair, a detection apparatus may determine that a foreign object is in proximity to one of the detection coils of the coil pair. In addition to the description of foreign object detection based on differential current, this disclosure provides several options for the design

and layout of detection coils to improve foreign object detection. Furthermore, this disclosure provides example circuit design options, layout design options, detection coil design options, and techniques to improve foreign object detection. For example, some of the provided design options may improve the accuracy of foreign object detection using unique detection coil designs and comparison of differential currents associated with different coil pairs in a detection apparatus. A movement of a wireless power reception apparatus during wireless power transfer may cause a change in impedance even in the absence of a foreign object. Advantageously, some techniques of this disclosure can distinguish the changes in impedance caused by a foreign object versus those caused by movement of the wireless power reception apparatus.

(27) A coil pair refers to two or more detection coils that can be concurrently excited (also referred to as energized) during an FOD period. The two or more detection coils may be located in respective foreign object detection zones (referred to as “detection zones” for brevity) in an interface space of the WPT system. In some implementations, the detection zones may be symmetrically located with regard to a primary magnetic field of the WPT system. For example, the detection zones may cover a scan area relative to a primary coil of a wireless power transmission apparatus. In some implementation, the detection zones may be symmetrically located with regard to a secondary coil of a wireless power reception apparatus. Alternatively, or additionally, the detection zones be located relative to both the primary coil and the secondary coil. A primary magnetic field refers to a magnetic field that is induced by a transmitter unit, such as the wireless power transmission apparatus of the WPT system. The detection coils may be in a detection apparatus (such as a detection mat or other device) constructed for use in the primary magnetic field. For example, a detection apparatus may be used in an interface space between a wireless power transmission apparatus and a wireless power reception apparatus. In some implementation, the detection apparatus can be a part of the wireless power transmission apparatus. In some implementations, the detection apparatus can be a part of the wireless power reception apparatus. In some implementation, the detection apparatus can be a stand-alone device that is independent of the wireless power transmission apparatus and wireless power reception apparatus.

(28) The coil pair may be coupled in a parallel circuit to a driver that concurrently excites the detection coils of the coil pair using a high frequency signal (such as 200 kHz or greater, as an example) during the FOD period. The differential current for that coil pair may be measured during the FOD period. A detection apparatus may have several such coil pairs and measure respective differential currents for the coil pairs. This disclosure includes example layouts of the detection coils in detection zones such that non-adjacent coil pairs can be excited at the same time in some implementations. Alternatively, each coil pair may be excited during different FOD periods to prevent cross-interference. By measuring the differential current associated with each coil pair and comparing the various differential currents for the coil pairs, the detection apparatus may determine whether a foreign object is present. And, in some implementations, the detection apparatus may determine a location of the foreign object in relation to one or more coil pairs.

(29) In some implementations, a differential current sensing apparatus may be used to determine the differential current associated with the coil pair. For example, differential current sensing apparatus may generate a detection voltage (or other detectable output value) having a magnitude that increases as the differential current of the coil pair increases, and vice versa. This disclosure describes an example differential current sensing apparatus that includes a magnetic core (such as a toroid) and a differential current sensing circuit. One leg of each detection coil of the coil pair may pass through the magnetic core in opposite directions. The currents associated with the detection coils generates a flux linkage with the magnetic core. When the currents of the detection coils are the same or similar, their flux linkages may cancel (or nearly cancel) each other such that the combined flux generated in the magnetic core is low. When the currents are dissimilar (indicative of a greater differential current), a higher combined flux is generated in the magnetic core.

(30) In some implementations, the differential current sensing circuit may include a sensor coil

wrapped around the magnetic core. The magnetic flux generated in the magnetic core may induce an electrical voltage signal in the differential current sensing circuit. That electrical signal may be rectified and filtered to produce a direct current voltage (referred to as a detection voltage) having a magnitude dependent on the differential current in the detection coils. Thus, for every coil pair, a detection voltage may represent the differential current in those coil pairs during the FOD period.

(31) In some implementations, a detection apparatus may include multiple coil pairs dispersed in detection zones. In some implementations, the detection zones may be non-overlapping (or only partially overlapping) and symmetrically located with regard to a primary magnetic field of the WPT system. For example, a circular detection area (also referred to as a scan area) may be divided into detection zones having a sector shape relative to a circular detection area. A detection apparatus may control which coil pair is excited during each FOD period. For example, the detection apparatus may excite each coil pair in a different FOD period. Alternatively, the detection apparatus may excite more than one coil pair that are in non-adjacent detection. Thus, the detection apparatus may prevent an adjacent detection coil from interfering with the differential current measurement for a particular coil pair in each FOD period.

(32) In some implementations, the detection apparatus may determine the detection voltages (corresponding to differential currents) associated with multiple coil pairs. Typically, a wireless power reception apparatus is large enough that it will simultaneously span multiple coil pairs. Conversely, a foreign object may be smaller than the wireless power reception apparatus. A foreign object may span only one coil pair or may span two coil pairs. The detection apparatus can distinguish between a movement of the wireless power reception apparatus versus introduction of a foreign object based on how many coil pairs have a change in differential current. For example, when the detection voltages for multiple coil pairs of a threshold quantity indicate a change in differential currents, the detection apparatus may determine that such change is a result of a movement of the wireless power reception apparatus. When the detection voltages for one coil pair (or below the threshold quantity) indicate a change in differential current, the detection apparatus may determine that such change is a result of the introduction of a foreign object. In some implementations, the detection apparatus may modify or offset the detection voltages for multiple coil pairs in response to determining that a movement of the wireless power reception apparatus has occurred. Thus, for a subsequent comparison of the detection voltages, the detection apparatus can adjust the detection voltages to account for a current location of the wireless power reception apparatus within the magnetic field of the WPT system. Thus, the accuracy of a subsequent FOD procedure can be improved by accounting for the normal impedance impact of the wireless power reception apparatus while still providing an accurate technique for detecting a foreign object introduced during wireless power transfer.

(33) This disclosure provides example designs for detection coils used in a detection apparatus. For example, the provided options for size, shape, construction, and location of the detection coils may improve the accuracy of the foreign object detection based on differential current. A size of the detection coils may be selected based on the disparate differences in sizes of a wireless power reception apparatus and a foreign object having a relatively smaller size compared to the wireless power reception apparatus. In some implementations, the size and shape of the detection coils of each coil pair may be uniform thereby normalizing the differences in impedance of the coils themselves. In some implementations, the detection coils may be constructed with a capacitance to increase the impedance of the detection coils for a power transfer frequency (such as 50 kHz) while offering low impedance when the detection coils are excited at a higher frequency (such as 200 kHz or greater).

(34) In some implementations, the coil pairs may be structured for use in a polygon or circle shaped scan area. For example, the detection coils may have a triangular or sector shape such that when they are placed in their respective non-overlapping detection zones, they form the polygon or circle shaped scan area. In some implementations, some coil pairs may be structured to prevent a narrow

detection area at the center of the scan area, while other coil pairs may be structured with a larger detection area to cover the center area of the scan area. For example, the detection coils of one or more coil pairs may have a triangular or sector shape with an additional portion to cover the center area, while the detection coils of other coil pairs may have a trapezoidal or annulus sector shape to fill the remaining portions of the polygon or circle shaped scan area.

(35) In some implementations, each detection coil may be constructed as a collection of smaller sub coils connected in series to form a single detection coil. The size of the smaller sub coils enables the detection coil to better capture the impedance impact of a smaller foreign object. Furthermore, in some implementations, the sub coils may be wound in opposite directions so that the primary magnetic field of the WPT system (such as during a wireless power transfer) will induce less or no voltage in the detection coil as a whole. Thus, the detection apparatus may stay in the primary magnetic field during wireless power transfer between the wireless power transmission apparatus and the wireless power reception apparatus. The voltage induced in the sub coils by the primary magnetic field may cancel each other out or reduce the overall voltage induced in the detection coil.

(36) This disclosure describes the use of detection coils in an FOD scan area. In some implementations, the size of the FOD scan area may be larger than a combined surface area of the power transfer coils of the WPT system. For example, a primary coil of a wireless power transmission apparatus may have a first diameter and a secondary coil of a wireless power reception apparatus may have a second diameter. An optimal placement of the wireless power reception apparatus and the wireless power transmission apparatus may be when the center of the primary coil and the secondary coil are perfectly aligned. However, such is not always the case in an actual operating environment. A WPT system may permit a misalignment tolerance where the secondary coil and the primary coil are misaligned but may still be capable of wireless power transfer. Meanwhile, a foreign object may be introduced in an area just outside the surface areas of a misaligned primary coil and secondary coil. Even in that location, the foreign object may absorb energy from the primary magnetic field during wireless power transfer and reduce the efficiency of the wireless power transfer or heat to dangerous temperatures. Thus, in some implementations, the FOD scan area may be a threshold size larger than a combined surface area of the power transfer coils taking into account the misalignment tolerance. In some implementations, the FOD scan area may have a diameter that is at least 10% larger than the larger of the primary coil diameter and the secondary coil diameter. In some implementations, the FOD scan area may have a diameter that is at least 10% larger than the larger power transfer coil plus the misalignment tolerance permitted by the WPT system.

(37) In some implementations, the FOD scan area may be dynamically determined based on characteristics of the primary coil, the secondary coil, their current alignment, or any combination thereof. For example, in a detection apparatus having multiple detection zones, the detection zones may be dynamically selected or disabled based on the current operating conditions. The dynamic FOD scan area size may be based at least a threshold size greater (such as 10% larger diameter or 20% larger radius) than the potential combined surface area or footprint of the power transfer coils plus the misalignment tolerance.

(38) In some implementations, a detection apparatus may include two parts such that a first part detects foreign objects in relation to a primary coil of a wireless power transmission apparatus and a second part detects foreign objects in relation to a secondary coil of a wireless power reception apparatus. Each of the parts may detect foreign objects in an FOD scan area that is at least a threshold size greater than a diameter of their respective power transfer coils. In such a configuration, even when the primary coil and the secondary coil are misaligned, the detection apparatus may detect foreign objects in a primary magnetic field relative to the primary coil and the secondary coil. Although described as two parts of a detection apparatus, in some implementations, each part may be constructed as separate detection mats. For example, a first detection mat may be

constructed, attached or integrated with a wireless power transmission apparatus and a second detection mat may be constructed, attached or integrated with a wireless power reception apparatus. (39) Particular implementations of the subject matter described in this disclosure can be implemented to realize one or more of the following potential advantages. A detection apparatus can detect foreign objects before wireless power transfer or in FOD periods during wireless power transfer. The techniques of this disclosure advantageously minimize impact of the detection apparatus on the wireless power transfer, and vice versa. Furthermore, the example detection coil design in this disclosure advantageously enables accurate foreign object detection using differential currents in a coil pair. The example layout of detection zones may enable fast and accurate detection of the foreign objects in an FOD scan area. The FOD scan area may be of a size sufficient to ensure that a foreign object at a peripheral of a wireless power transfer does not heat beyond a safe level.

(40) The following description is directed to certain implementations for the purposes of describing innovative aspects of this disclosure. However, a person having ordinary skill in the art will readily recognize that the teachings herein can be applied in a multitude of different ways. The described implementations can be implemented in any means, apparatus, system, or method for wireless power transfer.

(41) FIG. 1 shows a block diagram of an example wireless power transfer system **100**. The wireless power transfer system may include a wireless power transmission apparatus **102** and a wireless power reception apparatus **118**. The wireless power transmission apparatus includes a primary coil **110**. The primary coil **110** may be associated with a power signal generator **106**. The primary coil **110** may be a wire coil which transmits wireless power (which also may be referred to as wireless energy). The primary coil **110** may transmit wireless energy using inductive or magnetic resonant field. Together, the power signal generator and the primary coil may generate a primary magnetic field during wireless power transfer. The power signal generator **106** may include components (not shown) to provide power to the primary coil **110** causing the primary coil **110** to produce the wireless power signal. For example, the power signal generator **106** may include one or more switches, drivers, series capacitors, rectifiers or other components. The wireless power transmission apparatus **102** also may include a transmission controller **108** that controls the components of the power signal generator **106**. For example, the transmission controller **108** may determine an operating point (such as voltage or current) and control the power signal generator **106** according to the operating point.

(42) In some implementations, the power signal generator **106**, the transmission controller **108** and other components (not shown) may be collectively referred to as a power transmitter circuit. Some or all of the power transmitter circuit may be embodied as an integrated circuit (IC) that implements features of this disclosure for controlling and transmitting wireless power to one or more wireless power reception apparatuses. The transmission controller **108** may be implemented as a microcontroller, dedicated processor, integrated circuit, application specific integrated circuit (ASIC) or any other suitable electronic device.

(43) A power source **112** may provide power to the power transmitter circuit in the wireless power transmission apparatus **102**. The power source **112** may convert alternating current (AC) power to direct current (DC) power. For example, the power source **112** may include a converter that receives an AC power from an external power supply (such as a supply mains) and converts the AC power to a DC power used by the power signal generator **106**.

(44) In some implementations, a first communication unit **142** may be coupled to the components of the power signal generator **106** or the primary coil **110** to send or receive communications via the wireless power signal. The first communication unit **142** may include logic for controlling one or more switches and other components that cause transmission and reception of wireless signals via the wireless power signal. For example, the first communication unit **142** may include modulators or demodulators that convert information to modulated signals added to the wireless

power signal. In one example, the first communication unit **142** may convert data from the transmission controller **108** into a frequency shift key (FSK) modulated signal that is combined with the wireless power signal for a communication from the wireless power transmission apparatus **102** to the wireless power reception apparatus **118**. In another example, the first communication unit **142** may sense load modulated amplitude shift key (ASK) signals from the power signal generator **106** or the primary coil **110** and demodulate the ASK signals to obtain data that the first communication unit **142** provides to the transmission controller **108**.

(45) In some implementations, the wireless power transmission apparatus **102** may include a wireless communication interface **114**. The wireless communication interface **114** may be connected to a first communication coil **116** (which may be a coil or a loop antenna). The wireless communication interface **114** may include logic for controlling one or more switches and other components that cause transmission and reception of wireless communication signals via the first communication coil **116**. In some implementations, the wireless communication interface **114** may support short range radio frequency communication (such as Bluetooth™) or Near-Field Communication (NFC). NFC is a technology by which data transfer occurs on a carrier frequency of 13.56 MHz. The wireless communication unit **114** also may support any suitable communication protocol.

(46) The transmission controller **108** may detect the presence or proximity of a wireless power reception apparatus **118**. In some implementations, the presence or proximity of the wireless power reception apparatus **118** may be detected based on a load change in response to a periodic low power signal generated by the power signal generator **106** and the primary coil **110**. In some implementations, the presence or proximity of the wireless power reception apparatus **118** may happen during a periodic pinging process of the wireless communication interface **114** in the wireless power transmission apparatus **102**.

(47) The transmission controller **108** may control characteristics of wireless power that the wireless power transmission apparatus **102** provides to the wireless power reception apparatus **118**. After detecting the wireless power reception apparatus **118**, the transmission controller **108** may receive information from a wireless power reception apparatus **118**. For example, the transmission controller **108** may receive the information during a hand shaking process with the wireless power reception apparatus **118**. The information may include information about the wireless power reception apparatus **118** (such as a power rating, the manufacturer, the model, or parameters of the receiver when operating on a standard transmitter, among other examples). The transmission controller **108** may use this information to determine at least one operating control parameter (such as frequency, duty cycle, voltage, etc.) for wireless power it provides to the wireless power reception apparatus **118**. To configure the wireless power, the transmission controller **108** may modify the frequency, duty cycle, voltage or any other suitable characteristic of the power signal generator **106**.

(48) The wireless power reception apparatus **118** may include a secondary coil **120**, a rectifier **126**, and a receiver controller **128**. When the secondary coil **120** is aligned to the primary coil **110**, the secondary coil **120** may generate an induced voltage based on a received wireless power signal from the primary coil **110**. A capacitor may be in series between the secondary coil **120** and the rectifier **126**. The rectifier **126** may rectify the induced voltage and provide the induced voltage to a load **130**. In some implementations, the load **130** may be external to the wireless power reception apparatus **118** and coupled via electrical lines from the rectifier **126**.

(49) A receiver controller **128** may be connected to the rectifier **126** and a second communication unit **152**. The second communication unit **152** may be coupled to the components of the secondary coil **120** or the rectifier **126** to send or receive communications via the wireless power signal. The second communication unit **152** may include logic for controlling one or more switches and other components that cause transmission and reception of communication signals via the wireless power signals. For example, the second communication unit **152** may include modulators or demodulators

that convert information to ASK or FSK modulated signals. In one example, the second communication unit **152** may convert data from the receiver controller **128** into an ASK modulated signal that used to load modulate the wireless power signal for a communication from the wireless power reception apparatus **118** to the wireless power transmission apparatus **102**. In another example, the second communication unit **152** may sense FSK signals in the wireless power signal at the secondary coil **120** or the rectifier **126** and demodulate the FSK signals to obtain data that the second communication unit **152** provides to the receiver controller **128**.

(50) In some implementations, the wireless power reception apparatus **118** may include a wireless communication interface **132**. The wireless communication interface **132** may contain modulation and demodulation circuits to wirelessly communicate via a second communication coil **134** (which may be a coil or a loop antenna). Thus, the receiver controller **128** may wirelessly communicate with the transmission controller **108** via the wireless communication interface **132** and the wireless communication interface **114** using NFC communications or Bluetooth.

(51) In some traditional wireless power systems, a primary coil can transfer wireless energy to a secondary coil up to a rating predetermined by a wireless standard. For example, a low power wireless power signal may convey 5 Watts (5 W), 9 W, 12 W, or 15 W. A low power wireless power system may deliver up to 15 Watts of energy which is suitable for many electronic devices. Higher power wireless systems are being developed to support wireless power transmission to appliances or devices that require more power. For example, a high-power cordless kitchen transmitter may deliver power as high as 2.2 kW.

(52) An interface space **180** may demark a space between the wireless power transmission apparatus and the wireless power reception apparatus. For example, the interface space may include a surface of the wireless power transmission apparatus on which the wireless power reception apparatus may be placed. A distance between the primary coil **110** and the secondary coil may include a thickness of a surface in the interface space. During wireless power transfer, the primary coil **110** may induce a magnetic field (referred to as the primary magnetic field) through the interface space and into an operative environment in which the secondary coil is placed. Thus, the “operative environment” is defined by the primary magnetic field in the system, where the primary magnetic field of a primary coil **110** is detectably present and can detectably interact with the secondary coil or a foreign object **190** (shown as FO **190**). A foreign object **190** present in the operative environment of the WPT system may experience an increase in temperature due to interaction with the primary magnetic field. Therefore, when a foreign object is detected, the wireless power transmission apparatus may discontinue generating the primary magnetic field or otherwise prevent the wireless power transmission apparatus from transferring sufficient amounts of energy in the foreign object to cause the foreign object to heat beyond a safe level. Traditional techniques for detecting a foreign object may be based on a power loss accounting in which an amount of power received by the wireless power reception apparatus is compared with an amount of power output by the wireless power transmission apparatus and attribute a loss in power to a foreign object **190**. However, such techniques, when used alone, may be too slow, inaccurate, or insufficient in detecting the foreign object, particularly with higher power wireless power transfer systems.

(53) FIG. 2 shows a block diagram of an example detection apparatus in a wireless power transfer system **200**. The wireless power transfer system **200** includes a wireless power transmission apparatus **102** (with a primary coil **110**), an interface space **180**, and a wireless power reception apparatus **118** (with a secondary coil **120**) as described with reference to FIG. 1. For brevity, other components of the wireless power transmission apparatus **102** and the wireless power reception apparatus **118** are not shown in FIG. 2. A detection apparatus (such as the FOD mat **150** shown in FIG. 2, or variations thereof) may include a plurality of detection coils **170** capable of detecting the presence of a foreign object in accordance with some aspects of this disclosure. The detection coils **170** may be arranged in a pattern to cover an area at least a threshold greater than a relative

potential overlapping surface area of the primary coil **110** and the secondary coil **120**. In some implementations, the detection apparatus may include an FOD mat **150** and the detection coils may be constructed into or onto the FOD mat **150**. Although not shown in FIG. 2, in some implementations, the FOD mat may extend for a full area of the interface space. Alternatively, the FOD mat (and the quantity or configuration of the detection coils therein) may be sized based on a technical specification that defines the sizes of the primary coil **110**, the secondary coil, or both. While the example in FIG. 2 shows the detection apparatus deployed as a FOD mat **150** in or on a surface, in some implementations the detection apparatus may be deployed on or in any surface or construction in the space between the transmitter coil **110** and receiver coil **120**.

(54) The FOD mat **150** may be associated with a control unit **155**. In some implementations, the control unit **155** may communicate (shown as dashed arrow **157**) with the wireless power transmission apparatus **102** to enable or disable the wireless power transfer operations of the wireless power transmission apparatus **102** based on whether the control unit **155** detects a foreign object **190** in proximity to the detection coils **170**. In some implementations, the control unit **155** may communicate (not shown) with the wireless power reception apparatus **118** to enable or disable the wireless power transfer operations of the wireless power reception apparatus **118** based on whether the control unit **155** detects a foreign object **190** in proximity to the detection coils **170**. Although only one FOD mat **150** is shown in FIG. 2, in some implementations, two or more FOD mats may be deployed in a WPT system. For example, the control unit **155** may perform foreign object detection using FOD mats (not shown) located in relation to different primary coils of different wireless power transmission apparatuses in a stove top or other wireless power appliance. Alternatively, or additionally, one FOD mat may be located in association with the wireless power transmission apparatus and another FOD mat may be located in association with the wireless power reception apparatus. Each of the FOD mats may be connected to the same or different foreign object control units (performing the functions described with reference to the control unit **155** of FIG. 2).

(55) The technique by which a control unit **155** communicates with the wireless power transmission apparatus or the wireless power reception apparatus may vary. For example, the control unit **155** may have a wired communication link (not shown) to a transmission controller (not shown) of the wireless power transmission apparatus **102** or a receiver controller (not shown) of the wireless power reception apparatus **118**. In some implementations, the control unit **155** may communicate by a wireless communication link (not shown) with the wireless power transmission apparatus **102** or the wireless power reception apparatus **118**, or both. In some implementations, the control unit **155** may communicate with the wireless power transmission apparatus **102** or the wireless power reception apparatus **118** using a pin line or other control signal without a need for a communication protocol.

(56) The FOD mat **150** may be a flexible mat, a conformable mat, a rigid mat or a plug and play mat, a standalone mat, or combinations thereof. A substrate of the FOD mat **150** may be made of electrically insulating material. In some implementations, the FOD mat **150** may further include a mechanical wear resistant material to withstand movement of the wireless power reception apparatus over it (such as when the wireless power reception apparatus **118** is large appliance). In some implementations, the FOD mat **150** may further be designed for outdoor application and designed to withstand temperature, humidity and may be resistant to water ingress. The detection coils **170** may be disposed on a substrate of the FOD mat **150** or may be embedded in the substrate of the FOD mat **150** for user safety and aesthetics. In some other embodiments, the detection coils **170** may be printed, molded, woven, or additively manufactured on the substrate of the FOD mat **150**.

(57) The detection coils **170** may be operated in pairs. For example, a first detection coil **171** and a second detection coil **172** may form a coil pair. The control unit may excite the first detection coil **171** and the second detection coil **172** using a high frequency (higher than a frequency typically

used for the primary magnetic field). When present the foreign object may cause the first detection coil **171** to experience a different impedance compared to the second detection coil **172** (where no foreign object is present). By comparing the current drawn through the first detection coil **171** and the second detection coil **172**, the control unit may determine that the foreign object is present near the first detection coil **171** or the second detection coil **172**. The difference in current drawn by a coil pair may be referred to as a differential current. When the control unit determines that the foreign object is present based on the differential current, the control unit may cause the wireless power transmission apparatus to cease wireless power transmission.

(58) FIG. **3** shows a block diagram **300** of an example FOD mat configured to excite a pair of detection coils to measure a differential current. FIG. **3** shows a coil pair comprising a first detection coil **171** and a second detection coil **172**. The coil pair is connected in parallel to a driver **305**. Thus, when one of the detection coils in the coil pair is excited, so is the other detection coil. For brevity, the example in FIG. **3** is described as a coil pair having two detection coils. Each detection coil **171** and **172** may be a singular coil or may be constructed from two or more sub coils connected in series (as described further with reference to FIGS. **11** and **12**).

(59) A driver **305** may be operatively coupled to the coil pair (the first detection coil **171** and the second detection coil **172** in this example). The driver **305** may be configured to concurrently excite the detection coils **171** and **172** of the coil pair using an alternating current signal through coil connections **311** and **312**. In some implementations, the impedance values of the first detection coil **171** and the second detection coil **172** may be the same or similar when a foreign object **190** is not present. However, when the foreign object **190** is present, the foreign object **190** may cause a change in impedance to one of the detection coils **171** and **172** such that the first detection coil **171** has a first impedance value and second detection coil **172** has a second impedance value. The difference in impedance may cause an amount of current drawn through the coil connections **311** and **312** to differ. A differential current **350** may refer to a comparison of the current drawn through the coil connections **311** and **312**. When the foreign object **190** is not present and the impedance of the detection coils **171** and **172** are same or similar, the amount of current drawn through the coil connections **311** and **312** may be same or similar. Therefore, the differential current **350** may be a low value indicating little or no difference. Conversely, when the foreign object **190** is present near one of the first detection coil **171**, the impedance of that first detection coil **171** will change causing the differential current **350** to indicate a higher difference in the current drawn through the coil connections **311** and **312**.

(60) FIG. **4** shows a chart **400** with example magnitudes of differential currents. For example, FIG. **4** pictorially illustrates example magnitudes of differential currents **350A** and **350B** (as examples of the differential current **350** of FIG. **3**) and how the differential current may be used to determine whether a foreign object is present. When no foreign object is present (shown at graph **411**), a magnitude of the differential current **350A** may be lower than a differential current threshold level **420**. When a foreign object is present (shown at graph **412**), the magnitude of the differential current **350B** may be above the differential current threshold level. The differential current threshold level may be a configurable parameter based on a desired sensitivity of the detection apparatus.

(61) In the example shown in FIG. **4**, a foreign object is detected when the differential current is above a differential current threshold. In some implementations, the foreign object is detected based on an amount of change in the differential current. For example, the differential current may be higher during a baseline state and then decrease below a threshold amount when the foreign object is detected. The differential current may become greater or may become lesser (compared to a previous measurement or a baseline measurement) when a foreign object is present. Thus, in some implementations, the amount of change in differential current can indicate the presence of a foreign object. A change in the amount of the differential current may be compared with a delta threshold to determine whether the change is based on the introduction of a foreign object.

(62) FIG. 5 shows a block diagram of an example detection apparatus **500** based on a detection voltage induced by a differential current. The example detection apparatus may include detection coils arranged in pairs as described herein. For example, the example detection apparatus may include a pair of detection coils **171** and **172** (referred to as a coil pair) as described with reference to FIG. 3. For brevity, the driver and other components of the WPT system are not shown in FIG. 5. However, the detection apparatus includes a driver (not shown) configured to concurrently excite the coil pair during an FOD period. The presence (or lack thereof) of the foreign object **190** may cause a measurable differential current in the coil connections **311** and **312**. FIG. 5 provides one example of a differential current sensing apparatus that can be used to measure the differential current. The differential current sensing apparatus may include a magnetic core **510**, a differential current sensing circuit **501** and control unit **155**. The control unit **155** may be configured to generate a foreign object detection signal **580** based on the differential current in the coil connections **311** and **312**.

(63) When the currents on the coil connections **311** and **312** are passed through a magnetic core **510**, the difference in current generates a flux linkage in the magnetic core. The coil connections **311** and **312** are passed through the magnetic core in opposite directions so that an equal current in the coil connections **311** and **312** will generate a smaller flux linkage while differences in the current of the coil connections **311** and **312** generate a greater flux linkage. The flux linkage in the magnetic core **510** may induce a corresponding electrical signal in a sensor coil **520** wound around the magnetic core **510**. This induced electrical signal, under conditions of the magnetic core **510** not magnetically saturated, has an induced voltage **522** that is dependent on (such as related to or proportional to) the difference between current in the coil connections **311** and **312** and is representative of a measure of the differential current between the coil pair of detection coils **171** and **172**.

(64) The differential current sensing circuit **501** also may include a rectifier **530** that receives and rectifies the induced electrical voltage signal to generate a DC detection voltage **545** (referred to as detection voltage **545**). An optional filter **540** may filter the detection voltage before sending the detection voltage to the control unit **155**. In one example, the filter **540** is configured to filter out high frequency components from the measurement. In some implementations, the control unit **155** may include a comparator **560** configured to compare an absolute value of the detection voltage with a detection threshold **565**. Based on the comparison of the detection voltage with the detection threshold **565**, the control unit **155** may communicate a foreign object detection signal **580** or other control signal to a component of the WPT system (such as a wireless power transmission apparatus or a wireless power reception apparatus). For example, when the absolute value of the detection voltage is greater than the detection threshold **565**, the control unit **155** may indicate that a foreign object is present. Alternatively, the detection voltage in the presence of foreign object may be lower than the detection voltage when the foreign object is absent. Thus, in some implementations, a change in the absolute value of the detection voltage (from a previous or baseline measurement to the present measurement) may be compared with a delta threshold and a foreign object may be detected when the amount of change is greater than the delta threshold.

(65) The control unit **155** may include a component **555** to generate the absolute value (referred to as the magnitude) of the detection voltage.

(66) In some implementations, the control unit also may include calibration unit **550** configured to add or subtract an offset value (sometimes referred to as an “offset” for brevity) to the detection voltage. For example, the offset value may be based on a normal difference in impedance of the coil pair of detection coils **171** and **172** or may be based on a baseline measurement of the detection apparatus. In some implementations, the offset value may be determined during or after manufacturing of the detection coils **171** and **172**. Alternatively, or additionally, the offset value may be determined by the control unit or another test equipment (not shown) during a baseline measurement of the detection apparatus when no foreign object is present. For example, the offset

value may account for minor differences in impedance caused by other components of an FOD mat, a wireless power transmission apparatus, or a wireless power reception apparatus, depending on where the detection apparatus is installed. Alternatively, or additionally, the offset value may be based on a determination of the impedance difference as a result of a wireless power reception apparatus in the operating environment of the WPT system determined during a previous measurement.

(67) FIG. 6 shows a circuit diagram of an example electrical configuration **600** of a single coil pair. A power source **602** may provide power to a driver **690**. The driver **690** may be configured to concurrently excite a first detection coil **171** and a second detection **172** during an FOD period. As shown in FIG. 6, the detection coils **171** and **172** may be connected in parallel to the driver. Each of the detection coils **171** and **172** may have a capacitor **C1** or **C2**. In some implementations, the detection coils **171** and **172** may be referred to as inductors **L1** and **L2**, respectively. A driver controller **680** may be part of a control unit of the detection apparatus or may be implemented as a separate logical component of the detection apparatus. The driver may be a full bridge inverter. During an FOD period, in which the coil pair is excited to measure the differential current, the driver controller may control gates (A, B, C, and D) in the driver to cause the driver to generate an AC electrical signal to the detection coils **171** and **172**. In some implementations, the FOD period may occur before wireless power transfer between a wireless power transmission apparatus and a wireless power reception apparatus. Alternatively, or additionally, the FOD period may be associated with a measurement gap time period during which an ongoing wireless power transfer is suspended for the purposes of performing a foreign object detection procedure. FIG. 6 illustrates a configuration in which the detection coils **171** and **172** have coil connections passed through a magnetic core **510** of a differential current sensing apparatus. The coil connections may be passed through the magnetic core in opposite directions (as shown in FIG. 6).

(68) FIG. 7 shows a circuit diagram **700** of an example electrical configuration of multiple coil pairs. As described herein, a detection apparatus may utilize multiple coil pairs associated with corresponding detection zones. In some implementations, a detection apparatus may excite more than one coil pair at the same time, particularly when the coil pairs are associated with non-adjacent detection zones. In the circuit diagram, four coil pairs are shown. A first coil pair comprises first and second detection coils **171** and **172**. A second coil pair comprises third and fourth detection coils **173** and **174**. A third coil pair comprises fifth and sixth detection coils **175** and **176**. A fourth coil pair comprises seventh and eighth detection coils **177** and **178**. It should be apparent that the example of FIG. 7 is provided for pedagogical purposes and a detection apparatus may have any variety of coil pairs.

(69) The circuit diagram **700** includes a power source **602** configured to provide power to a driver unit **790**. The driver unit **790** may include one or more drivers. As shown in FIG. 7, the driver unit **790** may include two drivers, represented as half-bridge inverters. A first driver includes devices **712** and **714** forming a first half-bridge inverter. A second driver includes devices **716** and **718** forming a second half-bridge inverter. A driver controller **780** may control operation of the devices using control signals A, B, C, and D such that the first driver and the second driver may operate in a coordinated manner to excite the coil pairs at different times.

(70) Each coil pair may pass through a corresponding magnetic core **722**, **724**, **726** and **728**. Although illustrated as separate magnetic cores in FIG. 7, in some implementations, the quantity of magnetic cores may be reduced by having some coil pairs share a same magnetic core provided that only one coil pair per magnetic core is not excited during the FOD period. During a first FOD period, the driver controller **780** may cause a first coil pair (the detection coils **171** and **172**) to be excited and also may cause a third coil pair (the detection coils **175** and **176**) to be excited. The differential current sensing circuit (not shown) associated with each of the magnetic cores **722** and **724** may measure the detection voltage associated with the first coil pair and the third coil pair during the first FOD period. During a second FOD period, the driver controller **780** may cause a

second coil pair (the detection coils **173** and **174**) to be excited and also may cause a fourth coil pair (the detection coils **177** and **178**) to be excited. The differential current sensing circuit (not shown) associated with each of the magnetic cores **726** and **728** may measure the detection voltage associated with the second coil pair and the fourth coil pair during the second FOD period. Thus, the circuit diagram enables a driver controller **780** to manage excitation of multiple coil pairs by the driver unit **790** in accordance with a pattern.

(71) FIG. **8A** shows an example timing diagram **801** for a detection apparatus having multiple coil pairs. The timing diagram **801** shows the timing **860** of wireless power transfer in a WPT system and the timing **880** of a foreign object procedure of a detection apparatus. In some implementations, the foreign object procedure may be performed before a wireless power transfer **870**. Alternatively, or additionally, the foreign object procedure may be performed during a measurement gap **850** during which a wireless power transfer **805** is suspended to enable the foreign object detection procedure to occur with less interference caused by the primary magnetic field of the wireless power transfer. FIG. **8A** shows a sequential pattern in which a different coil pair **810**, **820**, **830**, and **840** is excited in each FOD period **851**, **852**, **853**, and **854**. For example, a first coil pair **810** may be excited and its differential current may be measured during a first FOD period **851**. Thereafter, the detection apparatus may discontinue exciting the first coil pair **810** and may excite a second coil pair **820** during a second FOD period **852** so that the differential current of the second coil pair **820** may be measured. The third coil pair **830** may be excited during a third FOD period **853**, and a fourth coil pair **840** may be excited during a fourth FOD period **854**. In this manner, each coil pair may be excited during a different FOD period so that its related differential current may be measured absent from interference of the other coil pairs.

(72) FIG. **8B** shows another example timing diagram **802** for a detection apparatus having multiple coil pairs. As with FIG. **8A**, the timing diagram **802** show the timing **860** of wireless power transfer in a WPT system. Different from FIG. **8A**, the detection apparatus may excite multiple coil pairs during each FOD period. For example, during the first FOD period **851**, the detection apparatus may excite the first coil pair **810** and the third coil pair **830** and measure their respective differential currents. During the second FOD period **852**, the detection apparatus may excite the second coil pair **820** and the fourth coil pair **840** and measure their respective differential currents. Thus, the measurement gap **850** may be shortened and the foreign object detection may be performed more quickly. In some implementations, the detection apparatus may use the circuit diagram described with reference to FIG. **7** to excite the various coil pairs. One consideration regarding which coil pair to excite during a same FOD period is the proximity or adjacency of the detection coils for the coil pairs. The examples in FIG. **8B** are based on a design in which the first coil pair **810** and the third coil pair **830** are in non-adjacent detection zones (and the second coil pair **820** and the fourth coil pair **840** are in non-adjacent detection zones). FIGS. **9A**, **9B** and **10** provide some example layouts using the non-adjacent coil pairs in the pattern described with reference to FIG. **8B**.

(73) FIG. **8C** shows another example timing diagram **803** for a detection apparatus having multiple coil pairs. As with FIG. **8B**, the timing diagram **803** in FIG. **8C** show the timing **860** of wireless power transfer in a WPT system. The coil arrangement used for FIG. **8C** may be similar to those described with reference to FIG. **8B**. FIG. **8C** shows an implementation in which the FOD periods may be interspersed during periods of wireless power transfer. For example, during a first FOD period **853**, the detection apparatus may excite the first coil pair **810** and the third coil pair **830** and measure their respective differential currents. Following the first FOD period **853**, the WPT system may have a period of wireless power transfer **871** before the next FOD period (shown as second FOD period **854**). During the second FOD period **854**, the detection apparatus may excite the second coil pair **820** and the fourth coil pair **840** and measure their respective differential currents. Another period of wireless power transfer **872** may follow the second FOD period **854**. The FOD periods **853** and **854** may be interspersed between periods of wireless power transfer **871** and **872**.

(74) FIG. 9A shows a diagram **900** of multiple coil pairs in associated detection zones. The coil pairs may be associated with detection zones. In some implementations, a detection mat may include several detection zones. The detection zones may be similar to geometric shapes such that the detection coils can cover a circle or polygon shaped scan area. The detection zones may define discrete regions of the detection mat. The detection zones may not be physically separated or isolated from one another. However, the detection zones may be electrically isolated from one another and further isolated based on the FOD periods in which they are used. Although geometric shapes are described in this disclosure for brevity, other shapes may be used in some implementations. The example in FIG. 9A illustrates triangular shaped detection coils **171**, **172**, **173**, **174**, **175**, **176**, **177** and **178**. For brevity, the detection coils **171**, **172**, **173**, **174**, **175**, **176**, **177** and **178** are referred to as L1, L2, L3, L4, L5, L6, L7 and L8, respectively. The detection coils may be triangular shaped and arranged in detection zones that form a polygon (such as an octagon in the example of FIG. 9A). In some implementations, the detection coils may have a sector shape such that they form a circle shaped FOD scan area. In some implementations, the coils may have other shapes of coils. The coils may be of uniform shape, size, construction material, and the like, resulting in a cost-effective design.

(75) FIG. 9A also shows the coil pairs **901** that are formed in this arrangement. A first coil pair **810** may include coils L1 and L2. As illustrated in FIG. 9A, the coils L1 and L2 may be located in detection zones that are symmetrically arranged with respect to a primary magnetic field (not shown) of the WPT system. A second coil pair **820** may include coils L3 and L4. A third coil pair **830** may include coils L5 and L6. A fourth coil pair **840** may include coils L7 and L8. The first coil pair **810** may be referred to as adjacent to the second coil pair **820** and the fourth coil pair **840** because at least one detection coil (the first detection coil **171**) is adjacent to a detection coil (such as the fourth detection coil **174** and the seventh detection coil **177**) of those coil pairs. Conversely, the first coil pair **810** and the third coil pair **830** may be non-adjacent coil pairs because the coils L1, L2, L5 and L6 of those coil pairs are non-adjacent. Thus, similar to the pattern described with reference to FIG. 8, the first coil pair **810** and the third coil pair **830** may be utilized during the same first FOD period. And the second coil pair **820** and the fourth coil pair **840** are non-adjacent coil pairs that may be utilized during the same second FOD period.

(76) FIG. 9B shows example detection of a foreign object using the example multiple coil pairs of FIG. 9A. In one example, a foreign object **960** may be located in proximity to the coil L3. When the coil pair of coils L3 and L4 are excited, those detection coils will draw different currents and the differential current may be indicative that the foreign object is located near one of those coils. In another example, a foreign object **970** may be located at the adjacent edges of coils L5 and L7. However, because the coil pairs that include those adjacent coils are excited during different FOD periods, the foreign object may still be detected. When the coil pair of coils L5 and L6 is excited, the foreign object will cause a differential current in that coil pair. When the coil pair of coils L7 and L8 are excited (in a different FOD period), the foreign object will cause a differential current in that coil pair.

(77) At the center area **980** of the FOD scan area, there may multiple detection coils in close proximity. Thus, when exciting multiple coil pairs there is a possibility of magnetic flux from one coil pair affecting that from another coil pair excited at the same time, resulting in a false determination of FO presence. Furthermore, the detection coils themselves may absorb energy from the primary magnetic field of the WPT system. The concentration of detection coils near the center **980** may result in heating caused by the primary magnetic field or interference with the wireless power transfer. Thus, in some implementations, the detection coils may be designed to reduce the concentration of coil material in the center area **980**.

(78) FIG. 10 shows another diagram **1000** of multiple coil pairs in associated detection zones. The layout diagram includes the detection coils **171**, **172**, **173**, **174**, **175**, **176**, **177** and **178** (referred to as L1, L2, L3, L4, L5, L6, L7 and L7, respectively) as described herein. Different from FIGS. 9A

and **9B**, the first and second detection coils **171** and **172** may have a different shape to address the concentration of coils in the center area **980** described with reference to FIG. **9B**. For example, the first coil pair (coils **L1** and **L2**) may be constructed to cover a triangular or a sector shaped portion of an FOD scan area and a center area of the FOD scan area. The other coils (**L3**, **L4**, **L5**, **L6**, **L7**, and **L8**) may cover trapezoidal or annulus sector shaped detection zones within the FOD scan area. It should be understood that the shapes of the detection coils in FIG. **10** are provided as an example. Other shapes may be possible. For example, more than one coil pair may have a unique design to cover the center area. In some implementations the center area may be covered by a coil pair in which the detection coils are the same shape. Alternatively, only one detection coil of the coil pair may cover the center area and the difference in impedance caused by the additional section of that detection coil may be accounted for by the control unit of the detection apparatus using an offset value.

(79) FIG. **11** shows an example line diagrams of detection coils. A first line diagram **1110** illustrates an example detection coil constructed from multiple sub coils connected in series. The sub coils together form a single detection coil. The first line diagram may be an example of the detection coils **L3**, **L4**, **L5**, **L6**, **L7**, and **L8** described with reference to FIG. **10**. The first line diagram shows eight (8) sub coils that are combined to form the detection coil. The use of sub coils enables the detection coil to be larger, reducing cost and complexity of the detection apparatus while still using smaller size sub coils for impedance response of a smaller foreign object. A second line diagram **1120** illustrates an example detection coil constructed from multiple sub coils connected in serial. The second line diagram may be an example of the detection coils **L1** and **L2** described with reference to FIG. **10**. The second line diagram shows ten (10) sub coils that are combined to form the detection coil. The second line diagram includes the same eight sub coils as the first line diagram and also includes two sub coils to cover a part of the center area of the FOD scan area.

(80) FIG. **12** shows an example line diagram of detection coils with features to reduce the effect of a power transfer coil field. The first line diagram and the second line diagram are the same as described with reference to FIG. **11**. FIG. **12** shows that the adjacent sub coils within each detection coil may be wound in opposite directions to reduce the effect of the primary magnetic field of the WPT system. For example, a first sub coil **1211** may be wound in a first direction (such as counterclockwise) and its adjacent sub coil **1212** may be wound in a second direction (such as clockwise). The layout of the sub coils may increase the likelihood that any voltages induced in the sub coils (from the primary magnetic field of the WPT system) nearly cancel out and the overall voltage across the detection coil terminals is small. This may protect some electronics in the detection apparatus, reducing power loss, or minimize heating of the sub coils, among other benefits. FIG. **12** also illustrates how adjacent sub coils can be wound in opposite directions in the second line diagram.

(81) Additionally, or alternatively, the detection coils illustrated in FIG. **12** may include capacitors **1230** in one or both terminals of the detection coils. The capacitors **1230** may be examples of the capacitors **C1**, **C2**, **C3**, **C4**, **C5**, **C6**, **C7**, and **C8** shown in FIGS. **6** and **7**, respectively. Some WPT systems are capable of transmitting wireless power using a frequency up to a maximum power frequency of 50 kHz. In accordance with some implementations of this disclosure, the detection coils may be excited using a higher frequency (such as 200 kHz or greater). The capacitors **1230** may reduce the effect of circulating currents due to the primary magnetic field (at or below 50 kHz) through the detection coil. The series capacitance of the capacitors **1230** may introduce a high impedance to the 50 kHz wireless power signal while still enabling the high frequency operation of the detection apparatus.

(82) FIG. **13** shows a diagram of multiple coil pairs capable of distinguishing between movement of a wireless power reception apparatus or a foreign object. The layout of FIG. **13** utilizes the example configuration of multiple coil pairs described with reference to FIG. **10**. A primary coil (not shown) may be positioned centered and underneath the detection zones covered by coils **L1**-

L8. A first example **1301** illustrates a secondary coil (shown by a single circle, for brevity) that moves from a first position **1310** to a second position **1320** over the array of detection coils. Because the secondary coil includes metallic and ferrite components, the secondary coil may cause a change in impedance of the detection coils. Thus, when the differential currents of the coil pairs are measured, the secondary coil itself may cause a differential current in various coil pairs. However, because the secondary coil is relatively large compared to the size of the detection coils, multiple coil pairs will measure the change in differential current. By comparison, a second example **1302** illustrates a foreign object **1330** introduced in the operating environment of the WPT system. The foreign object is relatively small compared to the size of the detection coils. Thus, perhaps only one or two coil pairs may measure a change in differential current as a result of the introduction of the foreign object. In some implementations, the size of the detection coils **L1-L8** (and their sub coils) may be selected based on a standardized size (or sizes) of a secondary coil that conforms to a technical specification. Similarly, the size of the detection coils **L1-L8** may be selected based on a potential size of foreign objects that are likely to be introduced into the operating environment. For example, a detection apparatus for use in a kitchen WPT system may size the detection coils appropriately to detect a spoon, fork, coin, key, tin can, or metal plate, among other examples. A detection apparatus for use in an EV WPT system may be sized appropriately to detect a wrench, aluminum can, gas tank, washer, nut, or screw, among other examples. A detection apparatus for use in a desktop WPT system may be sized appropriately to detect foreign object (such as a pen, keys, computer component, ring, or thumb drive, among other examples).

(83) FIG. **14** shows example charts **1400** comparing detection voltages induced by differential currents of coil pairs as a result of movement of a wireless power reception apparatus or a foreign object. In a first example **1410**, a detection apparatus may establish a baseline calibration of the detection voltages. For example, the detection apparatus may add offset values to the detection voltages for each coil pair based on a test calibration or baseline offset values. The detection voltages **1411**, **1412**, **1413** and **1414** show the detection voltages induced by the differential currents of multiple coil pairs (such as the coil pairs **810**, **820**, **830**, and **840**, described herein). The detection apparatus may determine that no foreign object is present based on the detection voltages. For example, the detection apparatus may compare each detection voltage with a detection threshold and determine, for each coil pair, that no foreign object is present. Alternatively, or additionally, the detection apparatus may compare an amount of change for each detection voltage with a delta threshold and determine that no foreign object is present when the amount of change is below the delta threshold. Based on a determination that no foreign object is present, the WPT system may begin wireless power transfer. The detection apparatus may determine a set of calibration values (which also may be referred to as offset values) based on the presence of the wireless power reception apparatus in the WPT system. The set of calibration values may be used to adjust the detection voltages for the coil pairs during a subsequent foreign object detection procedure.

(84) In a second example **1420** (such as a subsequent measurement), the wireless power reception apparatus may have moved from a previous location associated with a previous measurement. That movement may cause a change in the detection voltages for the coil pairs. For example, the detection voltages **1421**, **1422**, **1423** and **1424** show the detection voltages induced by the differential currents of the multiple coil pairs. In the second example, the detection voltages **1422**, **1423** and **1424** corresponding to the second, third, and fourth coil pair [has] have changed. The changes in the detection voltages **1422**, **1423**, and **1424** are circled at **1426**, **1427**, and **1428**, respectively, for the benefit of this description. Although illustrated as a decrease in detection voltages (circled at **1426**, **1427**, and **1428**) for pedagogical purposes, in some implementations the change may be an increase in detection voltages. Regardless of whether the change is an increase or a decrease, the detection apparatus may determine how many coil pairs have had a change in

detection voltage that is more than a delta threshold. The detection apparatus may determine that the changes are a result of a movement of the wireless power reception apparatus because more than a threshold quantity of coil pairs have had a change in their respective detection voltages. The threshold quantity may be a configurable parameter or may be predetermined. In some implementations, a change in three or more coil pairs may be deemed to be a result of a movement of the wireless power reception apparatus during wireless power transfer. Once the detection apparatus has determined that a change in the detection voltages is a result of a movement of the wireless power reception apparatus, the detection apparatus may adjust the set of offset values for the detection voltages to account for a new position of the wireless power reception apparatus. In some implementations, the offset values may be continuously adjusted during a wireless power transfer when the detection apparatus detects a movement of the wireless power reception apparatus. Once a wireless power transfer is completed or otherwise interrupted, the detection apparatus may reset the offset values back to a baseline calibration.

(85) In a third example **1430**, a foreign object may be introduced into the operating environment during wireless power transfer. As described with reference to FIG. **9**, the foreign object may impact the differential current of only one or two coil pairs. The detection voltages **1431**, **1432**, **1433** and **1434** show the detection voltages induced by the differential currents of the multiple coil pairs. In the third example, the detection voltage **1432** of the second coil pair has changed. The change in the detection voltage **1432** is circled at **1436** for the benefit of this description. Because the quantity of coil pairs having a change in detection voltage is less than a threshold quantity, the detection apparatus may determine that the change is a result of a foreign object introduced in proximity to the detection coils of the second coil pair. The detection apparatus may send a foreign object detection signal or control signal to the WPT system to stop the wireless power transfer to prevent the foreign object from becoming overheated.

(86) FIG. **15** shows a block diagram **1500** of an example detection apparatus configured to adapt detection voltages of coil pairs based on movement of a wireless power reception apparatus. The block diagram **1500** may describe features of a control unit in a detection apparatus with multiple coil pairs. The control unit may be configured to receive detection voltages **1511**, **1512**, **1513**, and **1514** from first, second, third, and fourth coil pairs, respectively. Each of the detection voltages may be obtained using different differential current sensing circuits associated with the coil pairs. The detection voltages **1511**, **1512**, **1513**, and **1514** may be adjusted using a set of offset values **1521**, **1522**, **1523**, and **1524**, respectively. Initially, the set of offset values (referred to, collectively, as offset values **1570**) may be based on a baseline calibration of the detection apparatus. The absolute values (magnitudes) of the detection voltages **1511**, **1512**, **1513**, and **1514** may be determined by components **1551**, **1552**, **1553**, and **1554**, respectively. A comparator **1560** may compare each adjusted detection voltage with a detection threshold **1565** to determine whether a foreign object is located near a detection coil of a particular coil pair. In some implementations, if any one of the detection voltages is above the detection threshold, the comparator may send a FOD signal **1580** indicating that a foreign object is detected. Alternatively, or additionally, the comparator may determine a quantity of the detection voltages that have changed from a previous iteration of the foreign object detection procedure. For example, the comparator may determine a change count indicating how many of the detection voltages have changed. If the change count is above a change threshold, the comparator may determine that the changes are a result of a movement of the wireless power reception apparatus and may adjust the offset values **1570** for a subsequent iteration of the foreign object detection procedure. Alternatively, if the change count is below a change threshold an FOD signal **1580** may be based solely on a comparison of each of the detection voltages to the detection threshold.

(87) FIG. **16** shows a pictorial diagram of example foreign object detection scan areas. In a first example **1601**, an FOD scan area **1632** may be based on a combined overlapping surface area of a primary coil **1610** and a secondary coil **1620**. In some implementations, the FOD scan area may be

larger than the combined overlapping surface area including any potential misalignment tolerance. For example, the first example shows several potential placements of the secondary coil within the misalignment tolerance. The FOD scan area may be at least a threshold size larger than the potential area of the primary coil **1610** and the secondary coil **1620** taking into account a maximum permitted misalignment of the primary coil and the secondary coil. In some implementations, the FOD scan area may be a circular area with a diameter at least 10% larger than the potential area of the primary coil and a maximally misaligned secondary coil. As an example, if a diameter of the primary coil is 210 millimeters (mm), a diameter of the secondary coil is 235 mm, and the permitted misalignment tolerance is 50 mm, the FOD scan area **1632** may be 358.5 mm ($2 * ((235/2) + (0.1 * (235/2)) + (50))$). The value of 358.5 mm for the FOD scan area **1632** may be based on the largest one of the primary coil and the secondary coil. In this example, the secondary coil is larger (235 mm). The formula includes multiplying a sum of the radius of the secondary coil ($235/2$) plus 10% of the radius of the secondary coil ($0.2 * 235$) plus the misalignment tolerance (50) by a value of 2 to determine a diameter of the FOD scan area **1632**. It should be understood that the example calculation is provided for pedagogical purposes and other calculations or formulas may be conducted to determine a size of the FOD scan area based on the size of the various power transfer coils.

(88) In a second example **1602**, the detection apparatus may utilize two FOD scan area—a first FOD scan area **1612** related to the primary coil and a second FOD scan area **1622** related to the secondary coil. The FOD scan areas **1612** and **1622** may be at least a threshold size larger than their respective power transfer coils (the primary coil and the secondary coil, respectively). For example, the first FOD scan area may have a diameter at least 10% larger than the diameter of the primary coil. The second FOD scan area **1622** may have a diameter at least 10% larger than diameter of the secondary coil. The opportunity to use two FOD scan areas enables detection of foreign objects relative to the power transfer coils regardless of any potential misalignment. In some implementations, a first detection apparatus may be utilized with the primary coil and a second detection apparatus may be used with the secondary coil.

(89) FIG. **17** shows a flowchart diagram of an example process **1700** for detecting a foreign metal object in accordance with some implementations. The operations of the process **1700** may be implemented by a detection apparatus as described herein. For example, the operations of process **1700** may be implemented by a detection apparatus described with reference to FIGS. **2-16**. The detection apparatus may have detection coils in any of the arrangements and constructions described herein. For brevity, the operations are described as performed by an apparatus. At block **1710**, the apparatus may concurrently excite at least a first detection coil and a second detection coil during a first FOD period. At block **1720**, the apparatus may detect, by a differential current sensing apparatus, a first differential current associated with the first detection coil and the second detection coil during the FOD period. At block **1730**, the apparatus may generate a foreign object detection signal based, at least in part, on the first differential current.

(90) FIG. **18** shows a block diagram of an example apparatus **1800** for use in wireless power transfer system. In some implementations, the apparatus **1800** may be a detection apparatus, such as any of the detection apparatuses described herein. The apparatus **1800** can include a processor **1802** (possibly including multiple processors, multiple cores, multiple nodes, or implementing multi-threading, etc.). The apparatus **1800** also can include a memory **1806**. The memory **1806** may be system memory or any one or more of the possible realizations of computer-readable media described herein. The apparatus **1800** also can include a bus **1811** (such as PCI, ISA, PCI-Express, HyperTransport®, InfiniBand®, NuBus®, AHB, AXI, etc.).

(91) The apparatus **1800** may include one or more controller(s) **1862** configured to manage excitation of multiple detection coils (such as a coil array **1864**). In some implementations, the controller(s) **1862** can be distributed within the processor **1802**, the memory **1806**, and the bus **1811**. The controller(s) **1862** may perform some or all of the operations described herein. For

example, the controller(s) **1862** may implement the features of a driver controller described herein. (92) The memory **1806** can include computer instructions executable by the processor **1802** to implement the functionality of the implementations described with reference to FIGS. **1-17**. Any one of these functionalities may be partially (or entirely) implemented in hardware or on the processor **1802**. For example, the functionality may be implemented with an application specific integrated circuit, in logic implemented in the processor **1802**, in a co-processor on a peripheral device or card, etc. Further, realizations may include fewer or additional components not illustrated in FIG. **18**. The processor **1802**, the memory **1806**, and the controller(s) **1862** may be coupled to the bus **1811**. Although illustrated as being coupled to the bus **1811**, the memory **1806** may be coupled to the processor **1802**.

(93) FIGS. **1-18** and the operations described herein are examples meant to aid in understanding example implementations and should not be used to limit the potential implementations or limit the scope of the claims. Some implementations may perform additional operations, fewer operations, operations in parallel or in a different order, and some operations differently.

(94) The foregoing disclosure provides illustration and description but is not intended to be exhaustive or to limit the aspects to the precise form disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the aspects. While the aspects of the disclosure have been described in terms of various examples, any combination of aspects from any of the examples is also within the scope of the disclosure. The examples in this disclosure are provided for pedagogical purposes. Alternatively, or in addition to the other examples described herein, examples include any combination of the following implementation options.

(95) One innovative aspect of the subject matter described in this disclosure can be implemented as a detection apparatus of a wireless power transfer (WPT) system. The detection apparatus may include a plurality of detection coils including at least a first detection coil and a second detection coil. The detection apparatus may include a first driver configured to concurrently excite the first detection coil and the second detection coil during a first foreign object detection (FOD) period. The detection apparatus may include a differential current sensing apparatus configured to detect a first differential current associated with the first detection coil and the second detection coil during the FOD period. The detection apparatus may include a control unit configured to generate a foreign object detection signal based on the first differential current.

(96) In some implementations, the detection apparatus may include a mat that includes the plurality of detection coils. The mat may be constructed for use in an interface space between a wireless power transmission apparatus and a wireless power reception apparatus.

(97) In some implementations, the first differential current is based on a difference between a first current of the first detection coil and a second current of the second detection coil during the FOD period.

(98) In some implementations, the differential current sensing apparatus includes at least a first magnetic core through which the first current of the first detection coil passes in a first direction and the second current of the second detection coil passes in a second direction such that the difference in the first current and second current generates a flux linkage in the first magnetic core representative of the first differential current.

(99) In some implementations, the differential current sensing apparatus further includes a first differential current sensing circuit that includes a first sensor coil wound around the first magnetic core. The first magnetic core may be configured to induce a first electrical signal in the first sensor coil based on the flux linkage. A magnitude of the induced voltage of the first electrical signal may be based on a magnitude of the first differential current.

(100) In some implementations, the differential current sensing circuit further includes a rectifier configured to rectify the first electrical signal to a direct current (DC) signal having a first detection voltage. The differential current sensing circuit may include a filter configured to stabilize the first

detection voltage of the DC signal.

(101) In some implementations, the control unit is configured to add or subtract a first offset to the first detection voltage to generate a first calibrated detection voltage.

(102) In some implementations, the control unit is configured to compare an absolute value of the first calibrated detection voltage with a detection threshold. The foreign object detection signal may indicate a foreign object is present when the first calibrated detection voltage is above the detection threshold. The foreign object detection signal may indicate that the foreign object is not present when the first calibrated detection voltage is below the detection threshold.

(103) In some implementations, the foreign object detection signal may indicate a foreign object is present when a change in the first calibrated detection voltage is above the delta threshold. The foreign object detection signal may indicate that the foreign object is not present when a change in the first calibrated detection voltage is below the delta threshold.

(104) In some implementations, the first detection coil and the second detection coil are positioned in respective detection zones in an interface space of the WPT system.

(105) In some implementations, the detection zones are symmetrically located with regard to either or both of a primary coil or a secondary coil of the WPT system.

(106) In some implementations, the plurality of detection coils includes one or more coil pairs, each coil pair having at least two associated detection coils connected in parallel to a driver. A first coil pair may include the first detection coil and the second detection coil.

(107) In some implementations, each coil pair is associated with at least two detection zones in an interface space of the WPT system.

(108) In some implementations, the one or more coil pairs include the first coil pair that includes the first detection coil and the second detection coil in a first detection zone and a second detection zone, respectively. In some implementations, the detection apparatus includes a second coil pair that includes a third detection coil and a fourth detection coil in a third detection zone and a fourth detection zone, respectively.

(109) In some implementations, the one or more coil pairs further include a third coil pair that includes a fifth detection coil and sixth detection coil in a fifth detection zone and a sixth detection zone, respectively.

(110) In some implementations, the one or more coil pairs further include a fourth coil pair that includes a seventh detection coil and an eighth detection coil in a seventh detection zone and an eighth detection zone, respectively.

(111) In some implementations, the detection apparatus may include a driver controller configured to control drivers associated with the one or more coil pairs such that the drivers concurrently excite the at least two associated detection coils of each coil pair during a respective FOD period. The differential current sensing apparatus may be configured to detect a corresponding differential current associated with each coil pair during the respective FOD period. The control unit may be configured to generate the foreign object detection signal based on the corresponding differential current associated with each coil pair.

(112) In some implementations, the driver controller is configured to cause the drivers to excite the one or more coil pairs in a consecutive pattern such that one coil pair is excited during each respective FOD period.

(113) In some implementations, the driver controller is configured to cause the drivers to excite the one or more coil pairs in a predetermined pattern such that two or more coil pairs are excited during each respective FOD period. The two or more coil pairs in each respective FOD period may be in non-adjacent detection zones.

(114) In some implementations, the differential current sensing apparatus includes a plurality of differential current sensing circuits configured to obtain a plurality of detection voltages, each detection voltage corresponding to a differential current of a respective coil pair. The plurality of detection voltages may include at least a first detection voltage of a magnitude that is based on the

first differential current of the first coil pair and a second detection voltage of magnitude dependent on the second differential current of the second coil pair.

(115) In some implementations, the control unit is configured to adjust the plurality of detection voltages based on a set of offset values. The set of offset values may include at least a first offset value for the first detection voltage and a second offset value for the second detection voltage.

(116) In some implementations, the set of offset values are initially based on a baseline measurement of impedance differences between the at least two associated detection coils for each coil pair.

(117) In some implementations, the control unit is configured to detect changes in the plurality of detection voltages over successive FOD periods for each detection voltage. The control unit may be configured determine a change count representing a quantity of the plurality of detection voltages that have changed over the successive FOD periods for each detection voltage.

(118) In some implementations, the control unit is configured to determine whether the changes in the plurality of detection voltages is caused by movement of a wireless power reception apparatus in the WPT system based on the change count.

(119) In some implementations, the control unit is configured to determine that the changes in the plurality of detection voltages is caused by movement of the wireless power reception apparatus when the change count is above or equal to a count threshold. The control unit may determine that the changes in the plurality of detection voltages is caused by a presence of a foreign object when the change count is below the count threshold.

(120) In some implementations, the control unit is configured to determine a ratio of the change count and a total count of the plurality of detection voltages. The control unit may be configured to determine that the changes in the plurality of detection voltages is caused by movement of the wireless power reception apparatus when the ratio is above or equal to a ratio threshold. The control unit may determine that the changes in the plurality of detection voltages is caused by a presence of a foreign object when the ratio is below the ratio threshold.

(121) In some implementations, the control unit is configured to modify the set of offset values based on the changes in the plurality of detection voltages when the control unit determines that the changes in the plurality of detection voltages is caused by movement of the wireless power reception apparatus in the WPT system.

(122) In some implementations, the control unit is configured to continuously calibrate the set of offset values to account for movement of the wireless power reception apparatus in the WPT system.

(123) In some implementations, the plurality of detection coils includes identically shaped detection coils.

(124) In some implementations, each of the at least two associated detection coils in each coil pair has a triangular shape or a sector shape such that when the plurality of detection coils are placed in corresponding detection zones. The plurality of detection coils may form a polygon or circle shaped FOD scan area.

(125) In some implementations, the plurality of detection coils includes one or more coil pairs having a first shape and one or more other coil pairs having a second shape.

(126) In some implementations, the first shape is substantially a trapezoidal shape or an annulus sector shape such that, when placed in corresponding detection zones, the one or more coil pairs of the first shape form part of a polygon or circle shaped FOD scan area excluding a center area of the FOD scan area. The second shape may be substantially a triangular or a sector shape combined with a polygonal shape or a sector shape such that, when placed in corresponding detection zones, the one or more other coil pairs of the second shape complete the polygon or circle shaped FOD scan area including the center area of the FOD scan area.

(127) In some implementations, the first detection coil and the second detection coil of the first coil pair are constructed to cover a triangular or a sector shaped portion of an FOD scan area and a

center area of the FOD scan area. Remaining ones of the plurality of detection coils may cover a trapezoidal or an annulus sector shaped portions, such that the plurality of detection coils form part of a polygon or circle shaped FOD scan area in which the center area is associated with the first coil pair.

(128) In some implementations, each detection coil of the plurality of detection coils is larger than a size of a reference foreign object.

(129) In some implementations, each detection coil of the plurality of detection coils is a size larger enough that a reference foreign object cannot simultaneously span more than two coil pairs.

(130) In some implementations, each detection coil of the plurality of detection coils is formed from sub coils connected in series to form a single detection coil.

(131) In some implementations, each sub coil is adjacent to one or more other sub coils to form an overall coverage area for the single detection coil.

(132) In some implementations, the single detection coil is constructed such that adjacent sub coils are wound in opposite directions thereby reducing an effect of a primary magnetic field of the WPT system on the single detection coil.

(133) In some implementations, each detection coil of the plurality of detection coils includes a series capacitance configured to reduce an effect of a primary magnetic field of the WPT system.

(134) In some implementations, the plurality of detection coils are constructed by litz wire.

(135) In some implementations, the plurality of detection coils are constructed on a printed circuit board (PCB).

(136) In some implementations, the plurality of detection coils, when excited, are operated at an FOD scan frequency that is above a power transfer frequency of the WPT system.

(137) In some implementations, the FOD scan frequency is 200 kilohertz (kHz) or greater. The power transfer frequency of the WPT system may be less than or equal to 50 kHz.

(138) In some implementations, the plurality of detection coils is arranged to cover a plurality of detection zones. The plurality of detection zones may form an FOD scan area that is at least a threshold size larger than a combined surface area of a primary coil and a secondary coil of the WPT system.

(139) In some implementations, the threshold size is determined based on a diameter of a larger one of the primary coil and the secondary coil.

(140) In some implementations, the threshold size is based on a diameter of a largest secondary coil supported by the WPT system.

(141) In some implementations, the threshold size is at least 10% greater than the diameter of a larger one of the primary coil and a largest secondary coil supported by the WPT system.

(142) In some implementations, the threshold size is further based on a maximum misalignment tolerance permitted between the primary coil and the secondary coil before wireless power transfer is disabled.

(143) In some implementations, the plurality of detection coils is arranged to cover a plurality of detection zones. The plurality of detection zones may be selectively enabled or disabled by the control unit to form a dynamic FOD scan area.

(144) In some implementations, the control unit is further configured to determine the dynamic FOD scan area based on potential combined surface area of a primary coil and a secondary coil of the WPT system.

(145) In some implementations, the dynamic FOD scan area large enough to cover a maximum misalignment tolerance permitted between the primary coil and the secondary coil as well as an additional adjacent surface area.

(146) In some implementations, the additional adjacent surface area causes the dynamic FOD scan area to be at least 10% larger in diameter than a largest diameter among the primary coil and the secondary coil supported by the WPT system including the maximum misalignment tolerance.

(147) In some implementations, a first subset of the plurality of detection coils are on a wireless

power transmission apparatus of the WPT system and a second subset of the plurality of detection coils are on the wireless power reception apparatus of the WPT system.

(148) Another innovative aspect of the subject matter described in this disclosure can be implemented as a detection apparatus of a wireless power transfer (WPT) system. The detection apparatus may include a plurality of detection coils arranged to form a foreign object detection (FOD) scan area that is at least a threshold size larger than a potential surface area of a plurality of power transfer coils of a wireless power transfer (WPT) system. The detection apparatus may include a control unit configured to generate a foreign object detection signal based on a detection of a foreign object in the FOD scan area.

(149) In some implementations, the plurality of power transfer coils include a primary coil of a wireless power transmission apparatus and a secondary coil of a wireless power reception apparatus. The threshold size may be determined based on a diameter of a larger one of the primary coil and the secondary coil.

(150) In some implementations, the threshold size is based on a diameter of a largest secondary coil supported by the WPT system.

(151) In some implementations, the threshold size is at least 10% greater than the diameter of a larger one of the primary coil and a largest secondary coil supported by the WPT system.

(152) In some implementations, the threshold size is further based on a maximum misalignment tolerance that the WPT system permits for misalignment of the primary coil and the secondary coil.

(153) In some implementations, a first subset of the plurality of detection coils are arranged on or in a wireless power transmission apparatus of the WPT system and a second subset of the plurality of detection coils are arranged on or in a wireless power reception apparatus of the WPT system.

(154) Another innovative aspect of the subject matter described in this disclosure can be implemented as a system. The system include a wireless power transmission apparatus including at least one primary coil and a first plurality of detection coils arranged to form a first foreign object detection (FOD) scan area that is at least a first size larger than a size of the primary coil. The system may include a wireless power reception apparatus including at least one secondary coil and a second plurality of detection coils arranged to form a second FOD scan area that is at least a second size larger than a size of the secondary coil.

(155) In some implementations, the first size of the first FOD scan area is based on a diameter of the primary coil. The second size of the second FOD scan area may be based on a diameter of the secondary coil.

(156) In some implementations, the first size is at least 10% larger than a diameter of the primary coil, and the second size is at least 10% larger than a diameter of the secondary coil.

(157) Another innovative aspect of the subject matter described in this disclosure can be implemented as a method of a detection apparatus for use in a wireless power transfer (WPT) system. The method may include concurrently exciting at least a first detection coil and a second detection coil during a first foreign object detection (FOD) period. The method may include detecting, by a differential current sensing apparatus, a first differential current associated with the first detection coil and the second detection coil during the FOD period. The method may include generating a foreign object detection signal based on the first differential current.

(158) In some implementations, the first differential current is based on a difference between a first current of the first detection coil and a second current of the second detection coil during the FOD period.

(159) In some implementations, detecting the first differential current includes passing the first current of the first detection coil in a first direction through a first magnetic core and the second current of the second detection coil in a second direction through the first magnetic core such that the difference in the first current and second current generates a flux linkage in the first magnetic core representative of the first differential current.

(160) In some implementations, the first magnetic core is configured to induce a first electrical

signal in a first sensor coil based on the flux linkage. A magnitude of the induced voltage of the first electrical signal may be based on a magnitude of the first differential current.

(161) In some implementations, the method may include rectifying the first electrical signal to a direct current (DC) signal having a first detection voltage. The method may include stabilizing, by a filter, the first detection voltage of the DC signal.

(162) In some implementations, the method may include adding or subtracting a first offset to the first detection voltage to generate a first calibrated detection voltage.

(163) In some implementations, the method may include comparing an absolute value of the first calibrated detection voltage with a detection threshold to determine the foreign object detection signal. The foreign object detection signal may indicate a foreign object is present when the first calibrated detection voltage is above the detection threshold. The foreign object detection signal may indicate that the foreign object is not present when the first calibrated detection voltage is below the detection threshold.

(164) In some implementations, the first detection coil and the second detection coil are positioned in respective detection zones in an interface space of the WPT system.

(165) In some implementations, the detection zones are symmetrically located with regard to either or both of a primary coil or a secondary coil of the WPT system.

(166) In some implementations, the detection apparatus includes a plurality of detection coils organized as one or more coil pairs, each coil pair having at least two associated detection coils connected in parallel to a driver. A first coil pair may include the first detection coil and the second detection coil.

(167) In some implementations, each coil pair is associated with at least two detection zones in an interface space of the WPT system.

(168) In some implementations, the method may include concurrently exciting the at least two associated detection coils of each coil pair during a respective FOD period. The method may include detecting a corresponding differential current associated with each coil pair during the respective FOD period. The method may include generating the foreign object detection signal based on the corresponding differential current associated with each coil pair.

(169) In some implementations, the method may include exciting the one or more coil pairs in a consecutive pattern such that one coil pair is excited during each respective FOD period.

(170) In some implementations, the method may include exciting the one or more coil pairs in a predetermined pattern such that two or more coil pairs are excited during each respective FOD period. The two or more coil pairs may be in each respective FOD period are in non-adjacent detection zones.

(171) In some implementations, the method may include obtaining, by a plurality of differential current sensing circuits, a plurality of detection voltages. Each detection voltage may correspond to a differential current of a respective coil pair. The plurality of detection voltages may include at least a first detection voltage of a magnitude that is based on the first differential current of the first coil pair and a second detection voltage of magnitude dependent on the second differential current of the second coil pair.

(172) In some implementations, the method may include adjusting the plurality of detection voltages based on a set of offset values. The set of offset values may include at least a first offset value for the first detection voltage and a second offset value for the second detection voltage.

(173) In some implementations, the set of offset values are initially based on a baseline measurement of impedance differences between the at least two associated detection coils for each coil pair.

(174) In some implementations, the method may include detecting changes in the plurality of detection voltages over successive FOD periods for each detection voltage. The method may include determining a change count representing a quantity of the plurality of detection voltages that have changed over the successive FOD periods for each detection voltage.

(175) In some implementations, the method may include determining whether the changes in the plurality of detection voltages is caused by movement of a wireless power reception apparatus in the WPT system based on the change count.

(176) In some implementations, the method may include determining that the changes in the plurality of detection voltages is caused by movement of the wireless power reception apparatus when the change count is above or equal to a count threshold. The method may include determining that the changes in the plurality of detection voltages is caused by a presence of a foreign object when the change count is below the count threshold.

(177) In some implementations, the method may include determining a ratio of the change count and a total count of the plurality of detection voltages. The method may include determining that the changes in the plurality of detection voltages is caused by movement of the wireless power reception apparatus when the ratio is above or equal to a ratio threshold. The method may include determining that the changes in the plurality of detection voltages is caused by a presence of a foreign object when the ratio is below the ratio threshold.

(178) In some implementations, the method may include modifying the set of offset values based on the changes in the plurality of detection voltages when the control unit determines that the changes in the plurality of detection voltages is caused by movement of the wireless power reception apparatus in the WPT system.

(179) In some implementations, the method may include continuously calibrating the set of offset values to account for movement of the wireless power reception apparatus in the WPT system.

(180) The figures, operations, and components described herein are examples meant to aid in understanding example implementations and should not be used to limit the potential implementations or limit the scope of the claims. Some implementations may perform additional operations, fewer operations, operations in parallel or in a different order, and some operations differently.

(181) As used herein, a phrase referring to “at least one of” or “one or more of” a list of items refers to any combination of those items, including single members. For example, “at least one of: a, b, or c” is intended to cover the possibilities of: a only, b only, c only, a combination of a and b, a combination of a and c, a combination of b and c, and a combination of a and b and c.

(182) The various illustrative components, logic, logical blocks, modules, circuits, operations and algorithm processes described in connection with the implementations disclosed herein may be implemented as electronic hardware, firmware, software, or combinations of hardware, firmware or software, including the structures disclosed in this specification and the structural equivalents thereof. The interchangeability of hardware, firmware and software has been described generally, in terms of functionality, and illustrated in the various illustrative components, blocks, modules, circuits and processes described above. Whether such functionality is implemented in hardware, firmware or software depends upon the particular application and design constraints imposed on the overall system.

(183) The hardware and data processing apparatus used to implement the various illustrative components, logics, logical blocks, modules and circuits described in connection with the aspects disclosed herein may be implemented or performed with a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device (PLD), discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, or any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, for example, a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some implementations, particular processes, operations and methods may be performed by circuitry that is specific to a given function.

(184) As described above, in some aspects of the subject matter described in this specification can be implemented as software. For example, various functions of components disclosed herein, or various blocks or steps of a method, operation, process or algorithm disclosed herein can be implemented as one or more modules of one or more computer programs. Such computer programs can include non-transitory processor-executable or computer-executable instructions encoded on one or more tangible processor-readable or computer-readable storage media for execution by, or to control the operation of, a data processing apparatus including the components of the devices described herein. By way of example, and not limitation, such storage media may include RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that may be used to store program code in the form of instructions or data structures. Combinations of the above should also be included within the scope of storage media.

(185) Various modifications to the implementations described in this disclosure may be readily apparent to persons having ordinary skill in the art, and the generic principles defined herein may be applied to other implementations without departing from the scope of this disclosure. Thus, the claims are not intended to be limited to the implementations shown herein but are to be accorded the widest scope consistent with this disclosure, the principles and the novel features disclosed herein.

(186) Additionally, various features that are described in this specification in the context of separate implementations also can be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation also can be implemented in multiple implementations separately or in any suitable subcombination. As such, although features may be described above as acting in particular combinations, and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

(187) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Further, the drawings may schematically depict one or more example processes in the form of a flowchart or flow diagram. However, other operations that are not depicted can be incorporated in the example processes that are schematically illustrated. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the illustrated operations. In some circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

Claims

1. A detection apparatus of a wireless power transfer (WPT) system, the detection apparatus comprising: a plurality of detection coils organized as pairs of detection coils that are symmetrically arranged in a foreign object detection (FOD) scan area of a wireless power transmission apparatus, wherein a first pair of detection coils includes at least a first detection coil and a second detection coil; a first driver configured to concurrently excite the first detection coil and the second detection coil; a differential current sensing apparatus configured to detect a first differential current associated with a difference between a first current of the first detection coil and a second current of the second detection coil while the first pair of detection coils is actively excited; and a control unit configured to detect a presence of a foreign object based, at least in part,

on the first differential current being above a differential current threshold.

2. The detection apparatus of claim 1, wherein the differential current sensing apparatus includes: at least a first magnetic core through which the first current of the first detection coil and the second current of the second detection coil are passed in opposite directions such that the difference in the first current and second current generates a flux linkage in the first magnetic core that is proportional to the first differential current; a first sensor coil wound around the first magnetic core, the flux linkage in the first magnetic core induces an induced voltage in the first sensor coil, wherein a magnitude of the induced voltage is proportional to a magnitude of the first differential current; and a rectifier and a filter configured to produce a direct current (DC) detection voltage based on the induced voltage of the first sensor coil.

3. The detection apparatus of claim 1, wherein: the differential current sensing apparatus is configured produce a direct current (DC) detection voltage that is proportional to the first differential current; and the control unit is configured to: add or subtract a first offset to the DC detection voltage to generate a first calibrated detection voltage, compare an absolute value of the first calibrated detection voltage with a detection threshold, and detect the presence of the foreign object when the first calibrated detection voltage is above the detection threshold.

4. The detection apparatus of claim 1, further comprising: the differential current sensing apparatus configured to obtain a plurality of detection voltages, each detection voltage corresponding to a differential current of a respective pair of detection coils, wherein the plurality of detection voltages includes at least a first detection voltage of a magnitude that is proportional to the first differential current of the first pair of detection coils and a second detection voltage of magnitude that is proportional to a second differential current of a second pair of detection coils.

5. The detection apparatus of claim 4, further comprising: a driver controller configured to cause one or more drivers to excite the pairs of detection coils according to: a first pattern in which pairs of detection coils are sequentially excited one pair at a time, a second pattern in which two or more pairs of detection coils that are not adjacent to each other are excited at a same time, or a third pattern in which a first set of two or more pairs of detection coils that are not adjacent to each other are excited during a first FOD slot and a second set of two or more pairs of detection coils that are not adjacent to each other are excited during a second FOD slot.

6. The detection apparatus of claim 4, wherein the control unit is configured to: detect changes in the plurality of detection voltages over successive measurements for each detection voltage; and counting, as a change count, a quantity of the plurality of detection voltages that exceed a detection threshold since a previous measurement for each detection voltage.

7. The detection apparatus of claim 6, wherein the control unit is configured to determine, based on the change count, whether the changes in the plurality of detection voltages are caused by movement of a wireless power reception apparatus or the presence of the foreign object.

8. The detection apparatus of claim 6, wherein the control unit is configured to: determine that the changes in the plurality of detection voltages are caused by movement of a wireless power reception apparatus when the change count is above or equal to a count threshold, and determine that the changes in the plurality of detection voltages are caused by the presence of the foreign object when the change count is below the count threshold.

9. The detection apparatus of claim 6, wherein the control unit is configured to: determine a ratio of the change count and a total count of the plurality of detection voltages, determine that the changes in the plurality of detection voltages are caused by movement of a wireless power reception apparatus when the ratio is above or equal to a ratio threshold, and determine that the changes in the plurality of detection voltages are caused by the presence of the foreign object when the ratio is below the ratio threshold.

10. The detection apparatus of claim 4, wherein the control unit is configured to adjust the plurality of detection voltages based on a set of offset values, the set of offset values including at least a first offset value for the first detection voltage and a second offset value for the second detection

voltage, wherein the set of offset values are initially based on a baseline measurement of impedance differences between at least two associated detection coils for each pair.

11. The detection apparatus of claim 10, wherein the control unit is configured to recalibrate the set of offset values to account for movement of a wireless power reception apparatus based on changes in the plurality of detection voltages when the control unit determines that the changes are caused by the movement of the wireless power reception apparatus.

12. A method of a detection apparatus, the method comprising: organizing a plurality of detection coils into pairs of detection coils that are symmetrically arranged in a foreign object detection (FOD) scan area of a wireless power transmission apparatus wherein a first pair of detection coils includes at least a first detection coil and a second detection coil; concurrently exciting the first detection coil and the second detection coil; detecting a first differential current associated with a difference between a first current of the first detection coil and a second current of the second detection coil while the first pair of detection coils is actively excited; and detecting, by a control unit, a presence of a foreign object based, at least in part, on the first differential current being above a differential current threshold.

13. The method of claim 12, further comprising: producing a direct current (DC) detection voltage that is proportional to the first differential current; adding or subtracting a first offset to the DC detection voltage to generate a first calibrated detection voltage; comparing an absolute value of the first calibrated detection voltage with a detection threshold; and detecting the presence of the foreign object when the first calibrated detection voltage is above the detection threshold.

14. The method of claim 12, further comprising: obtaining a plurality of detection voltages, each detection voltage corresponding to a differential current of a respective pair of detection coils, wherein the plurality of detection voltages includes at least a first detection voltage of a magnitude that is proportional to the first differential current of the first pair of detection coils and a second detection voltage of magnitude that is proportional to a second differential current of a second pair of detection coils.

15. The method of claim 14, further comprising: causing one or more drivers to excite the pairs of detection coils according to: a first pattern in which pairs of detection coils are sequentially excited one pair at a time, a second pattern in which two or more pairs of detection coils that are not adjacent to each other are excited at a same time, or a third pattern in which a first set of two or more pairs of detection coils that are not adjacent to each other are excited during a first FOD slot and a second set of two or more pairs of detection coils that are not adjacent to each other are excited during a second FOD slot.

16. The method of claim 14, further comprising: detecting changes in the plurality of detection voltages over successive measurements for each detection voltage; counting, as a change count, a quantity of the plurality of detection voltages that exceed a detection threshold since a previous measurement for each detection voltage; and determining, based on the change count, whether the changes in the plurality of detection voltages is caused by movement of a wireless power reception apparatus or the presence of the foreign object.

17. The method of claim 16, further comprising: determining that the changes in the plurality of detection voltages are caused by the movement of the wireless power reception apparatus when the change count is above or equal to a count threshold, and determining that the changes in the plurality of detection voltages are caused by the presence of the foreign object when the change count is below the count threshold.

18. The method of claim 16, further comprising: determining a ratio of the change count and a total count of the plurality of detection voltages; determining that the changes in the plurality of detection voltages are caused by the movement of the wireless power reception apparatus when the ratio is above or equal to a ratio threshold; and determining that the changes in the plurality of detection voltages are caused by the presence of the foreign object when the ratio is below the ratio threshold.

19. The method of claim 14, further comprising: adjusting the plurality of detection voltages based on a set of offset values, the set of offset values including at least a first offset value for the first detection voltage and a second offset value for the second detection voltage; and recalibrating the set of offset values to account for movement of a wireless power reception apparatus based on changes in the plurality of detection voltages when the control unit determines that the changes are caused by the movement of the wireless power reception apparatus.

20. A foreign object detection (FOD) system, comprising: an FOD mat including multiple pairs of detection coils, wherein two detection coils of each pair are symmetrically arranged in an FOD scan area, and the FOD scan area is at least 10% larger in diameter than a combined surface area associated with a primary coil of a wireless power transmission apparatus and a secondary coil of a wireless power reception apparatus; a driver unit configured to excite pairs of detection coils at an excitation frequency that is above a power transfer frequency, the excitation frequency being 200 kilohertz (kHz) or greater, wherein, for a first pair, the driver unit concurrently excites the two detection coils in the first pair, and wherein the pairs of detection coils are excited according to a pattern that prevents adjacent pairs from being concurrently excited; a sensor configured to generate a plurality of direct current (DC) detection voltages that are proportional to corresponding differential currents associated with differences between the two detection coils in the first pair when the first pair is actively excited; and a control unit configured to: calibrate the plurality of DC detection voltages based on a set of offset values, wherein the set of offset values are initially based on a baseline measurement of impedance differences between the two detection coils for each pair, count, as a change count, a quantity of the calibrated plurality of DC detection voltages that exceed a detection threshold since a previous measurement for each detection voltage, detect a presence of a foreign object when the change count is below a count threshold, detect a movement of the wireless power reception apparatus when the change count is above or equal to the count threshold, and recalibrate the set of offset values based on changes in the plurality of DC detection voltages when the control unit detects the movement of the wireless power reception apparatus.
